Seminar 7 - 18.11.2021 - GR312CD - Algaed.

**T. Cayley-Hamilton** $A \in M_n(\mathbb{R})$

$P_A(\lambda) = \det(A - \lambda I_n) = (-1)^n \cdot \prod_{i=1}^{k} (\lambda - \lambda_i)^{n_i}$

$\lambda_1 \to n_1$ ; $\lambda_k \to n_k$ ; $n_1 + \dots + n_k = n$

$P_A(\lambda) = a_0 + a_1\lambda + \dots + a_{n-1}\lambda^{n-1} + (-1)^n \cdot \lambda^n$.

$\boxed{P_A(A) = O_n}$ i.e. $a_0 \cdot I_n + a_1 \cdot A + \dots + (-1)^n \cdot A^n = O_n$.

Ex 1 $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \boxed{A^2 - (\operatorname{tr} A)\cdot A + (\det A)\cdot I_2 = O_2}$.

$P_A(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = (a-\lambda)(d-\lambda) - bc \Rightarrow$

$\Rightarrow P_A(\lambda) = \lambda^2 - \lambda(\underbrace{a+d}_{\operatorname{Tr} A}) + \underbrace{ad - bc}_{\det A} \Rightarrow P_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\cdot\lambda + \det A$.

Cayley-Haw.; $n=2 \Rightarrow \boxed{P_A(A) = O_2} \Rightarrow A^2 - (\operatorname{tr} A)\cdot A + (\det A) I_2 = O_2$.

Ex 2 $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$; $A^n = ?$ fol. Cayley-Hamilton.

$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} -\lambda & 1 & 0 \\ 1 & 1-\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = \lambda^2(1-\lambda) + \lambda + \lambda =$

$= \lambda(-\lambda^2 + \lambda + 2) =$

$= -\lambda(\lambda - 2)(\lambda + 1)$

$\boxed{X^n = P_A(X)\cdot Q(X) + aX^2 + bX + c}$.

$P_A(X) = -X(X-2)(X+1)$.

$A^n = \underbrace{P_A(A)}_{O_3}\cdot Q(A) + \underbrace{a\cdot A^2 + b\cdot A + c\cdot I_3}$ $\Rightarrow \boxed{A^n = aA^2 + bA + c\cdot I_3}$.

? a, b, c.

$X = 0 \Rightarrow 0^n = c \Rightarrow \boxed{c = 0}$.

$X = -1 \Rightarrow (-1)^n = \underbrace{P_A(-1)}_{0}\cdot Q(-1) + a\cdot(-1)^2 + b\cdot(-1) + c \Rightarrow \boxed{(-1)^n = a - b}$

$X = 2 \Rightarrow 2^n = 4a + 2b + c \Rightarrow \boxed{2a + b = 2^{n-1}}$

$\Rightarrow a, b \Rightarrow \boxed{A^n}$.

Ex3. $A=\begin{pmatrix}0&2&1\\-1&1&0\\0&0&2\end{pmatrix}$. $A^{-1}=?$ fol. polin. caract.

$$P_A(\lambda)=\det(A-\lambda I_3)=\begin{vmatrix}-\lambda&2&1\\-1&1-\lambda&0\\0&0&2-\lambda\end{vmatrix}=-\lambda(1-\lambda)(2-\lambda)+2(2-\lambda)=$$
$$=(2-\lambda)\left[2-\lambda+\lambda^2\right].$$

$$P_A(\lambda)=4-2\lambda-2\lambda+\lambda^2+2\lambda^2-\lambda^3=4-4\lambda+3\lambda^2-\lambda^3.$$

$\boxed{P_A(A)=O_3}\Rightarrow -A^3+3A^2-4A+4\cdot I_3=O_3\cdot 1\cdot A^{-1}\Rightarrow$

$$\Rightarrow 4A^{-1}-4I_3+3A-A^2=O_3\Rightarrow$$

$$\boxed{A^{-1}=I_3-\frac{3}{4}A+\frac{1}{4}A^2}$$

## Diagonalizarea matricilor

**Def.** $A\in M_n(\mathbb{R})$ e **diagonalizabilă** dc $\exists P\in M_n(\mathbb{R})$ nesingulară ($\det P\neq 0$). a.î. $P^{-1}AP=D$. unde $D=$ mat diag $=\begin{pmatrix}d_1&&0\\&\ddots&\\0&&d_n\end{pmatrix}$.

**T.** $V$ sp. vect/$K$; $\dim V=n$; $f\in \mathrm{End}_K V$ ($f:V\to V$ apl lin)

$$P_f(x)=P_{A_f}(x)=(-1)^n\cdot\prod_{i=1}^{k}(\lambda-\lambda_i)^{n_i}\Rightarrow \boxed{\text{UAE}}:$$

a) $f$ diagonalizabilă.

b) $\exists B$ bază pt $V$ formată din vect proprii ai lui $f$.

c) $V_{\lambda_1}\oplus V_{\lambda_2}\oplus\dots\oplus V_{\lambda_k}=V$

d) $\boxed{\dim V_{\lambda_i}=g_i=n_i\ ;\ \forall i=\overline{1,k}}$

### Algoritm de diagonalizare

$\sigma(A)=\{\lambda_1,\dots,\lambda_k\}$.

1) $P_A(\lambda)=0=\det(A-\lambda I_n)\Rightarrow \lambda_1,\dots,\lambda_k\to n_1,\dots,n_k$.

2) pt fiec val proprie $\lambda\Rightarrow V_\lambda=\{\vec{v}\in\mathbb{R}^n / A\vec{v}=\lambda\vec{v}\}$.

3) $\boxed{\dim V_{\lambda_i}=n_i\ ;\ \forall i=\overline{1,k}}$ $\Rightarrow$ (DA) $\Rightarrow$ A se diag $\Rightarrow$ PASUL 4.

(NU) $\Rightarrow$ A nu se diag $\Rightarrow$ STOP.

4) $A=PDP^{-1}$; $D=\begin{pmatrix}\lambda_1&&&&0\\&\overbrace{\lambda_1}^{n_1}&&&\\&&\ddots&&\\&&&\lambda_k&\\0&&&&\lambda_k\end{pmatrix}$; $P=$ mat. având drept col vectori proprii lin ind.

x].

$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ $A^n = ?$

? diagonalizati.

$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = \begin{vmatrix} 3-\lambda & 1 & 1 \\ 3-\lambda & 1-\lambda & 1 \\ 3-\lambda & 1 & 1-\lambda \end{vmatrix} =$

$= (3-\lambda) \cdot \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = (3-\lambda) \cdot \begin{vmatrix} 1 & 0 & 0 \\ 1 & -\lambda & 0 \\ 1 & 0 & -\lambda \end{vmatrix} = \lambda^2(3-\lambda).$

$\Rightarrow \lambda_1 = 0 ; n_1 = 2.$

$\lambda_2 = 3 ; n_2 = 1.$

$\boxed{\lambda_1 = 0} \Rightarrow \vec{v} = ? ; \vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \underline{A\vec{v} = \lambda_1 \vec{v}} \Leftrightarrow \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$

$x + y + z = 0 \Rightarrow \underline{z = -x - y} \cdot \Rightarrow \vec{v} = \begin{pmatrix} x \\ y \\ -x-y \end{pmatrix} = x \underbrace{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}}_{v_1} + y \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{v_2}.$

$V_{\lambda_1} = \mathrm{Sp}\{v_1, v_2\} \cdot \}$ $v_1, v_2$ liniar ind. $\} \Rightarrow \{v_1, v_2\}$ bază pt $V_{\lambda_1}$.

$\boxed{\dim V_{\lambda_1} = 2.}$

$\boxed{\lambda_2 = 3} \Rightarrow ? \vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in A\vec{v} = \lambda_2 \cdot \vec{v} \Rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 3 \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$

$x + y + z = 3x = 3y = 3z \Rightarrow x = y = z \cdot \Rightarrow$

$\Rightarrow \vec{v} = x \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{v_3} \Rightarrow V_{\lambda_2} = \mathrm{Sp}\{\underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{v_3}\} \cdot ) \boxed{\dim V_{\lambda_2} = 1.}$

$\dim V_{\lambda_1} = 2 = n_1 \cdot$
$\dim V_{\lambda_2} = 1 = n_2 \cdot$ $\Big] \Rightarrow A$ e diag. $A = PDP^{-1}$ ; $D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

$P = (v_1 | v_2 | v_3) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & -1 & 1 \end{pmatrix} \Rightarrow \boxed{A = PDP^{-1}}.$

$A^n = PD^nP^{-1} = P \cdot \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3^n \end{pmatrix} P^{-1} \Rightarrow \ldots$ ◻